SEQUENCE 5

Term review and remediation

16–20 November 2026

5.1 Individual review

Method · Instructions. Work without the course notes for 25 minutes, then check with the methods in the booklet. For each question, note: solved alone, solved with help, or to redo. Justify the answers.

Example 1 · Review of sequences 1 to 4.

1. Reduce $90/126$ to lowest terms. Prove that the sum of two odd integers is even.
2. With $A(-1,2)$ and $B(3,-2)$, calculate $\overrightarrow{AB}$, AB and the midpoint of $[AB]$ in an orthonormal frame.
3. Determine an equation of (AB); does $C(5,-4)$ belong to this line?
4. Calculate $[-2,4]\cap(1,7]$ and $[-2,4]\cup(1,7]$.
5. Solve $|x-1|\leq 4$ and explain why $1/3\notin\mathbb{D}$.

Solution. 1. $\gcd(90,126)=18$, so $90/126=5/7$. If $u=2k+1$ and $v=2\ell+1$, then $u+v=2(k+\ell+1)$ is even.
2. $\overrightarrow{AB}=(4,-4)$; $AB=4\sqrt{2}$; midpoint $I(1,0)$.
3. $y=-x+1$. For C, $-4=-5+1$: it belongs to (AB).
4. Intersection $(1,4]$; union $[-2,7]$.
5. $x\in[-3,5]$. Writing $1/3=a/10^n$ would require 10^n to be divisible by 3, which is impossible.

Teacher note. Possible marking out of 20: 4 points per question. Assess calculation skill, choice of method and quality of justification separately.

5.2 Correcting an error and revisiting a method

Example 2 · Analysis of student work. Correct: (a) "$\overrightarrow{AB} = (x_A - x_B, y_A - y_B)$"; (b) "The zero vector directs every line"; (c) "$|x - 2| \leq 3$ gives $x \in [2, 3]$"; (d) "$1/3 \approx 0.333$, so $1/3$ is decimal".

Solution. (a) Subtract the origin from the endpoint: $(x_B - x_A, y_B - y_A)$. (b) A direction vector must be nonzero. (c) The distance to 2 is at most 3: $x \in [-1, 5]$. (d) 0.333 differs from $1/3$: $3 \times 0.333 = 0.999 \neq 1$. A decimal approximation does not determine the nature of the approximated number.

Example 3 · Remediation track.

1. Reduce $72/108$, then $\frac{3}{4} - \frac{1}{6}$.
2. With $P(1, 1)$ and $Q(5, 3)$, determine the midpoint, a direction vector and an equation of (PQ).
3. Determine $(-\infty, 3] \cap [-1, +\infty)$, then solve $|x + 2| < 1$.

Solution. 1. $2/3$; $7/12$. 2. Midpoint $(3, 2)$, direction vector $(2, 1)$, equation $x - 2y + 1 = 0$. 3. Intersection $[-1, 3]$; solution of the inequality $(-3, -1)$.

5.3 Applying and self-assessing

Example 4 · Synthesis problem. Two displacements are represented by $\vec{u}(6,4)$ and $\vec{v}(9,6)$.

1. Show that they are collinear and find the coefficient relating $\vec{v}$ to $\vec{u}$.
2. The starting point is $A(-2,1)$. Determine the endpoints B and C associated with the displacements $\vec{u}$ and $\vec{v}$.
3. Find an equation of the line through these three points.
4. A point $M(x,y)$ of this line must satisfy $|x-4| \leq 3$. Give the interval of possible abscissae.

Solution. 1. $\vec{v} = \frac{3}{2}\vec{u}$. 2. $B(4,5)$ and $C(7,7)$. 3. A direction vector is $(3,2)$; the equation is $2x-3y+7=0$, satisfied by the three points. 4. $|x-4| \leq 3 \iff x \in [1,7]$.

Key point · My review. My priority: .....

The mistake to avoid: .....

The method to reuse: .....

An exercise to redo without help: .....

Teacher note. Validate the remediation with a similar exercise done without help after a delay. Orally, ask for a sentence linking the calculation to the geometric conclusion.